

MATERIAL TEST REPORT

HFW STEEL PIPE GENERAL INFORMATION																	
ITEM	O.D.(in)	Wall Thickness (in)	Length (ft)	Specification(s) or Reference and Grade	Coating	End Finish	NDT Weld Seam	NDT Pipe Body	Drift	Visual and Dimension Control	Flattening Test	Min Collapse Pressure	Hydrostatic Test		Heat Treatment	Workorder	Date
													Pressure (Psi)	Hold Time (sec)			
1	10,750	0,500	38,00-47,00	API 5L 46th ed X52M PSL2	Zero VOC Clear Mill Coating	30 Beveled End	UT: OK N 10: 0.125in through hole	API 5L - E3.2 Pipe End Inspection - Welded Pipe	N/A	OK	OK	N/A	4200	5	Weld Seam Annealed Min 1598° F	2503W018591	3/19/2025

ITEM	Heat No	Lot No	Chemical Analysis															
				C (%)	Mn (%)	P (%)	S (%)	Si (%)	Al (%)	Cu (%)	Ni (%)	Cr (%)	Mo (%)	V (%)	Nb (%)	Ti (%)	B (%)	Ceq
			Max	0.220	1.400	0.025	0.015	0.450		0.200	0.200	0.100	0.150	0.050	0.050	0.040		0.250
			Min															
			Ceq (sum)	C (%)	Mn % / 20			Si % / 30		Cu % / 20	Ni % / 60	Cr % / 20	Mo % / 15	V % / 10	Nb + V + Ti <= 0.15%		5 X B%	
1	SDX-A405923	N/A	Product	0.078	0.610	0.010	0.002	0.230	0.014	0.124	0.046	0.069	0.019	0.000	0.023	0.014	0.0007	0.131
			Skelp	0.073	0.610	0.010	0.002	0.232	0.014	0.125	0.046	0.069	0.018	0.000	0.024	0.014	0.0007	0.126
			Heat	0.060	0.610	0.012	0.003	0.220	0.018	0.140	0.050	0.060	0.020	0.001	0.025	0.015	0.0003	0.112

			Mechanical Properties																
			Transverse Weld								Transverse Body								
			Seam Weld, 8" overall, 2.250" reduction area, 2" gauge length. Flat specimen		Impact Test at 32° F (ft/lb)						Pipe Body, 10.250" over all length, 2.250" length reduction area, 2" gauge length, T180, Flat specimen			Impact Test at 32° F (ft/lb)/ 3T90					
			Tensile Strength (PSI)	Vickers Hardness HV10	Sample Size:	Full					Yield Strength (PSI)	Tensile Strength (PSI)	Elongation (%)	Vickers Hardness HV10	Sample Size:	1/2			
Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear				Average Impact Energy	Average Shear (%)	Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear						Average Impact Energy	Average Shear (%)		
ITEM	Heat No	Lot No	Max								76,900								
			Min	66,700		20	20	20			52,200	66,700			20	20	20		
			1	SDX-A405923	N/A	Product	74,471	171	394.9 / 90	379.2 / 90	416.5 / 90	397	90	62,203	71,735	41	168	426.1 / 90	406.2 / 90

Heat No	Impact Test at -20° F (ft/lb)					
	Sample Size:	Full				
		Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy	Average Shear (%)
	20	20	20			
SDX-A405923	388 / 90	389.1 / 100	388.1 / 90	388	93	

Heat No	Impact Test at -20° F (ft/lb)/ 3T90					
	Sample Size:	1/2				
		Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy	Average Shear (%)
	20	20	20			
SDX-A405923	413.7 / 90	429 / 90	456.4 / 90	433	90	

SOP-24-F25
Inspection Certificate - Line Pipe
ISO 10474:1991

 BORUSAN
PIPE
4949 Borusan Road
Baytown, Texas 77523

MATERIAL TEST REPORT

Coil Source: Steel Dynamics (1945 Airport Rd, Columbus, MS 39701)
Melted and Manufactured in USA

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WE CERTIFY THE ABOVE MATERIAL HAS BEEN MANUFACTURED, PROCESSED, SAMPLED, TESTED, AND INSPECTED TO MEET THE REQUIREMENTS OF THE STANDARD IN REFERENCE.
DETERMINATION OF MECHANICAL TESTING ARE ACCURATE AND TRACEABLE TO NIST AND CONFORM TO APPLICABLE TESTING METHODS.
CONFORMS TO NACE MR-0175 FOR HARDNESS ONLY AS IT APPLIES TO API PRODUCTS. MAX HARDNESS DOES NOT EXCEED RC22.
NOT VALID WITH INADEQUATE USE OF PIPE.
THIS INSPECTION CERTIFICATE IS IN ACCORDANCE WITH API 5L 46TH EDITION.

Borusan Pipe

Todd Reeves /


VP OF QUALITY ASSURANCE

Revision : 4
Revision date : 2/12/2024

Reviewed by: Technical Manager and Sr. Metallurgist
Approved by: VP of QA

Data Collector:
QA Coordinator: